

Nand2Tetris Project 3

HDL Code + Chip Simulation Screenshots

Updated document including the newly provided PC and Register simulator screenshots.

Component	HDL Code	Screenshot
Bit	Included	Included
Register	Included	Included
RAM8	Included	Included
RAM64	Included	Included
RAM512	Included	Included
RAM4K	Included	Included
RAM16K	Included	Included
PC	Included	Included

1. Bit.hdl

HDL Source Code

```
01 // This file is part of www.nand2tetris.org
02 // and the book "The Elements of Computing Systems"
03 // by Nisan and Schocken, MIT Press.
04 // File name: projects/3/a/Bit.hdl
05 /**
06  * 1-bit register:
07  * If load is asserted, the register's value is set to in;
08  * Otherwise, the register maintains its current value:
09  * if (load(t)) out(t+1) = in(t), else out(t+1) = out(t)
10  */
11 CHIP Bit {
12   IN in, load;
13   OUT out;
14
15   PARTS:
16   Mux(a=prevOut, b=in, sel=load, out=muxOut);
17   DFF(in=muxOut, out=prevOut, out=out);
18
19 }
```

File: Bit(1).hdl | Chip: Bit

1. Bit — Simulation Screenshot

HDL ☐ --LBSt

Project 3

Bit

```
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/3/a/Bit.hdl
5 /**
6  * 1-bit register:
7  * If load is asserted, the register's value is set to in;
8  * Otherwise, the register maintains its current value:
9  * if (load(t)) out(t+1) = in(t), else out(t+1) = out(t)
10 */
11 CHIP Bit {
12     IN in, load;
13     OUT out;
14
15     PARTS:
16     Mux(a=prevOut, b=in, sel=load, out=muxOut);
17     DFF(in=muxOut, out=prevOut, out=out);
18
19 }
```

Chip Bit

Eval Reset Clock: 0

Input pins

in 0

load 0

Output pins

out 0

Internal pins

prevOut 0

muxOut 0

Provided screenshot from the HDL/chip simulator.

2. Register.hdl

HDL Source Code

```
01 // This file is part of www.nand2tetris.org
02 // and the book "The Elements of Computing Systems"
03 // by Nisan and Schocken, MIT Press.
04 // File name: projects/3/a/Register.hdl
05 /**
06  * 16-bit register:
07  * If load is asserted, the register's value is set to in;
08  * Otherwise, the register maintains its current value:
09  * if (load(t)) out(t+1) = int(t), else out(t+1) = out(t)
10  */
11 CHIP Register {
12   IN in[16], load;
13   OUT out[16];
14
15   PARTS:
16   Bit(in=in[0], load=load, out=out[0]);
17   Bit(in=in[1], load=load, out=out[1]);
18   Bit(in=in[2], load=load, out=out[2]);
19   Bit(in=in[3], load=load, out=out[3]);
20   Bit(in=in[4], load=load, out=out[4]);
21   Bit(in=in[5], load=load, out=out[5]);
22   Bit(in=in[6], load=load, out=out[6]);
23   Bit(in=in[7], load=load, out=out[7]);
24   Bit(in=in[8], load=load, out=out[8]);
25   Bit(in=in[9], load=load, out=out[9]);
26   Bit(in=in[10], load=load, out=out[10]);
27   Bit(in=in[11], load=load, out=out[11]);
28   Bit(in=in[12], load=load, out=out[12]);
29   Bit(in=in[13], load=load, out=out[13]);
30   Bit(in=in[14], load=load, out=out[14]);
31   Bit(in=in[15], load=load, out=out[15]);
32 }
```

File: Register(1).hdl | Chip: Register

2. Register — Simulation Screenshot

HDL ☐ --LBSt Project 3 Register

```
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/3/a/Register.hdl
5 /**
6  * 16-bit register:
7  * If load is asserted, the register's value is set to in;
8  * Otherwise, the register maintains its current value:
9  * if (load(t)) out(t+1) = int(t), else out(t+1) = out(t)
10 */
11 CHIP Register {
12     IN in[16], load;
13     OUT out[16];
14
15     PARTS:
16     Bit(in=in[0], load=load, out=out[0]);
17     Bit(in=in[1], load=load, out=out[1]);
18     Bit(in=in[2], load=load, out=out[2]);
19     Bit(in=in[3], load=load, out=out[3]);
20     Bit(in=in[4], load=load, out=out[4]);
21     Bit(in=in[5], load=load, out=out[5]);
22     Bit(in=in[6], load=load, out=out[6]);
23     Bit(in=in[7], load=load, out=out[7]);
24     Bit(in=in[8], load=load, out=out[8]);
25     Bit(in=in[9], load=load, out=out[9]);
26     Bit(in=in[10], load=load, out=out[10]);
27     Bit(in=in[11], load=load, out=out[11]);
28     Bit(in=in[12], load=load, out=out[12]);
29     Bit(in=in[13], load=load, out=out[13]);
30     Bit(in=in[14], load=load, out=out[14]);
31     Bit(in=in[15], load=load, out=out[15]);
32 }
```

Chip Register Eval Reset Clock: 0

Input pins

in 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

load 0

Output pins

out 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

Provided screenshot from the HDL/chip simulator.

3. RAM8.hdl

HDL Source Code

```
01 // This file is part of www.nand2tetris.org
02 // and the book "The Elements of Computing Systems"
03 // by Nisan and Schocken, MIT Press.
04 // File name: projects/3/a/RAM8.hdl
05 /**
06  * Memory of eight 16-bit registers.
07  * If load is asserted, the value of the register selected by
08  * address is set to in; Otherwise, the value does not change.
09  * The value of the selected register is emitted by out.
10  */
11 CHIP RAM8 {
12   IN in[16], load, address[3];
13   OUT out[16];
14
15   PARTS:
16   DMux8Way(in=load, sel=address, a=load0, b=load1, c=load2, d=load3, e=load4, f=load5, g=load6,
17   h=load7);
18   Register(in=in, load=load0, out=reg0);
19   Register(in=in, load=load1, out=reg1);
20   Register(in=in, load=load2, out=reg2);
21   Register(in=in, load=load3, out=reg3);
22   Register(in=in, load=load4, out=reg4);
23   Register(in=in, load=load5, out=reg5);
24   Register(in=in, load=load6, out=reg6);
25   Register(in=in, load=load7, out=reg7);
26
27   Mux8Way16(a=reg0, b=reg1, c=reg2, d=reg3, e=reg4, f=reg5, g=reg6, h=reg7, sel=address, out=out);
28 }
```

File: RAM8(1).hdl | Chip: RAM8

3. RAM8 — Simulation Screenshot

HDL

--LBST

Project 3

RAM8

```
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/3/a/RAM8.hdl
5 /**
6  * Memory of eight 16-bit registers.
7  * If load is asserted, the value of the register selected by
8  * address is set to in; Otherwise, the value does not change.
9  * The value of the selected register is emitted by out.
10 */
11 CHIP RAM8 {
12     IN in[16], load, address[3];
13     OUT out[16];
14
15     PARTS:
16     DMux8Way(in=load, sel=address, a=load0, b=load1, c=load2, d=load3, e=load4, f=load5, g=load6, h=load7);
17
18     Register(in=in, load=load0, out=reg0);
19     Register(in=in, load=load1, out=reg1);
20     Register(in=in, load=load2, out=reg2);
21     Register(in=in, load=load3, out=reg3);
22     Register(in=in, load=load4, out=reg4);
23     Register(in=in, load=load5, out=reg5);
24     Register(in=in, load=load6, out=reg6);
25     Register(in=in, load=load7, out=reg7);
26
27     Mux8Way16(a=reg0, b=reg1, c=reg2, d=reg3, e=reg4, f=reg5, g=reg6, h=reg7, sel=address, out=out);
28 }
```

Chip RAM8

Eval Reset Clock: 0

Input pins

in 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

load 0

address 0 0 0 dec

Output pins

out 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

Internal pins

load0 0

load1 0

load2 0

load3 0

load4 0

load5 0

load6 0

load7 0

reg0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

reg1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

reg2 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

reg3 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

reg4 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

reg5 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

reg6 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

reg7 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

Visualization

Register: 0

Register: 0

Register: 0

Register: 0

Register: 0

Register: 0

Register: 0

Register: 0

Provided screenshot from the HDL/chip simulator.

4. RAM64.hdl

HDL Source Code

```
01 // This file is part of www.nand2tetris.org
02 // and the book "The Elements of Computing Systems"
03 // by Nisan and Schocken, MIT Press.
04 // File name: projects/3/a/RAM64.hdl
05 /**
06 * Memory of sixty four 16-bit registers.
07 * If load is asserted, the value of the register selected by
08 * address is set to in; Otherwise, the value does not change.
09 * The value of the selected register is emitted by out.
10 */
11 CHIP RAM64 {
12   IN in[16], load, address[6];
13   OUT out[16];
14
15   PARTS:
16   DMux8Way(in=load, sel=address[3..5], a=load0, b=load1, c=load2, d=load3, e=load4, f=load5, g=load6,
17   h=load7);
18   RAM8(in=in, load=load0, address=address[0..2], out=ram0);
19   RAM8(in=in, load=load1, address=address[0..2], out=ram1);
20   RAM8(in=in, load=load2, address=address[0..2], out=ram2);
21   RAM8(in=in, load=load3, address=address[0..2], out=ram3);
22   RAM8(in=in, load=load4, address=address[0..2], out=ram4);
23   RAM8(in=in, load=load5, address=address[0..2], out=ram5);
24   RAM8(in=in, load=load6, address=address[0..2], out=ram6);
25   RAM8(in=in, load=load7, address=address[0..2], out=ram7);
26
27   Mux8Way16(a=ram0, b=ram1, c=ram2, d=ram3, e=ram4, f=ram5, g=ram6, h=ram7, sel=address[3..5],
28   out=out);
29 }
```

File: RAM64(1).hdl | Chip: RAM64

4. RAM64 — Simulation Screenshot

The screenshot shows the HDL/Chip Simulator interface. The left pane displays the HDL code for the RAM64 chip, and the right pane shows the chip's internal structure and visualization.

HDL Code (Left Pane):

```
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/3/a/RAM64.hdl
5 /**
6  * Memory of sixty four 16-bit registers.
7  * If load is asserted, the value of the register selected by
8  * address is set to in; Otherwise, the value does not change.
9  * The value of the selected register is emitted by out.
10 */
11 CHIP RAM64 {
12     IN in[16], load, address[6];
13     OUT out[16];
14
15     PARTS:
16     DMux8Way(in=load, sel=address[3..5], a=load0, b=load1, c=load2, d=load3, e=load4, f=load5, g=load6, h=load7);
17
18     RAM8(in=in, load=load0, address=address[0..2], out=ram0);
19     RAM8(in=in, load=load1, address=address[0..2], out=ram1);
20     RAM8(in=in, load=load2, address=address[0..2], out=ram2);
21     RAM8(in=in, load=load3, address=address[0..2], out=ram3);
22     RAM8(in=in, load=load4, address=address[0..2], out=ram4);
23     RAM8(in=in, load=load5, address=address[0..2], out=ram5);
24     RAM8(in=in, load=load6, address=address[0..2], out=ram6);
25     RAM8(in=in, load=load7, address=address[0..2], out=ram7);
26
27     Mux8Way16(a=ram0, b=ram1, c=ram2, d=ram3, e=ram4, f=ram5, g=ram6, h=ram7, sel=address[3..5], out
28
29 }
```

Chip RAM64 (Right Pane):

The right pane shows the chip's internal structure and visualization. It includes input pins (in, load, address), output pins (out), internal pins (load0 through load7), and a visualization of the RAM8 components (ram0 through ram7).

Input pins:

- in: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
- load: 0
- address: 0 0 0 0 0 0

Output pins:

- out: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Internal pins:

- load0: 0
- load1: 0
- load2: 0
- load3: 0
- load4: 0
- load5: 0
- load6: 0
- load7: 0

Visualization:

The visualization shows the RAM8 components (ram0 through ram7) and their internal state. The RAM8 components are shown as a table with columns for Address and Value.

RAM8	Addr	dec
0	0	0
1	0	0
2	0	0
3	0	0
4	0	0

Provided screenshot from the HDL/chip simulator.

5. RAM512.hdl

HDL Source Code

```
01 // This file is part of www.nand2tetris.org
02 // and the book "The Elements of Computing Systems"
03 // by Nisan and Schocken, MIT Press.
04 // File name: projects/3/b/RAM512.hdl
05 /**
06  * Memory of 512 16-bit registers.
07  * If load is asserted, the value of the register selected by
08  * address is set to in; Otherwise, the value does not change.
09  * The value of the selected register is emitted by out.
10  */
11 CHIP RAM512 {
12   IN in[16], load, address[9];
13   OUT out[16];
14
15   PARTS:
16   DMux8Way(in=load, sel=address[6..8], a=load0, b=load1, c=load2, d=load3, e=load4, f=load5, g=load6,
17   h=load7);
18   RAM64(in=in, load=load0, address=address[0..5], out=ram0);
19   RAM64(in=in, load=load1, address=address[0..5], out=ram1);
20   RAM64(in=in, load=load2, address=address[0..5], out=ram2);
21   RAM64(in=in, load=load3, address=address[0..5], out=ram3);
22   RAM64(in=in, load=load4, address=address[0..5], out=ram4);
23   RAM64(in=in, load=load5, address=address[0..5], out=ram5);
24   RAM64(in=in, load=load6, address=address[0..5], out=ram6);
25   RAM64(in=in, load=load7, address=address[0..5], out=ram7);
26
27   Mux8Way16(a=ram0, b=ram1, c=ram2, d=ram3, e=ram4, f=ram5, g=ram6, h=ram7, sel=address[6..8],
28   out=out);
29 }
```

File: RAM512(1).hdl | Chip: RAM512

5. RAM512 — Simulation Screenshot

HDL ⌵ --LBS1
Project 3
RAM512
Eval Reset Clock: 0

```

1 file is part of www.nand2tetris.org
2 e book "The Elements of Computing Systems"
3 an and Schocken, MIT Press.
4 ame: projects/5/b/RAM512.hdl
5
6 of 512 16-bit registers.
7 d is asserted, the value of the register selected by
8 s is set to in; Otherwise, the value does not change.
9 lue of the selected register is emitted by out.
10
11 12 {
12 [16], load, address[9];
13 ut[16];
14
15 :
16 Way(in=load, sel=address[6..8], a=load0, b=load1, c=load2, d=load3, e=load4, f=load5, g=load6, h=load7)
17
18 (in=in, load=load0, address=address[0..5], out=ram0);
19 (in=in, load=load1, address=address[0..5], out=ram1);
20 (in=in, load=load2, address=address[0..5], out=ram2);
21 (in=in, load=load3, address=address[0..5], out=ram3);
22 (in=in, load=load4, address=address[0..5], out=ram4);
23 (in=in, load=load5, address=address[0..5], out=ram5);
24 (in=in, load=load6, address=address[0..5], out=ram6);
25 (in=in, load=load7, address=address[0..5], out=ram7);
26
27 ay16(a=ram0, b=ram1, c=ram2, d=ram3, e=ram4, f=ram5, g=ram6, h=ram7, sel=address[6..8], out=out);
28

```

Chip RAM512

Input pins

in 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

load 0

address 0 0 0 0 0 0 0 0 0 dec

Output pins

out 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

Internal pins

load0 0

load1 0

load2 0

load3 0

load4 0

load5 0

load6 0

load7 0

ram0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

ram1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

ram2 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

ram3 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

ram4 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

ram5 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

ram6 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

ram7 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

Visualization

RAM64	Addr	Value
0		0
1		0
2		0
3		0
4		0

Provided screenshot from the HDL/chip simulator.

6. RAM4K.hdl

HDL Source Code

```
01 // This file is part of www.nand2tetris.org
02 // and the book "The Elements of Computing Systems"
03 // by Nisan and Schocken, MIT Press.
04 // File name: projects/3/b/RAM4K.hdl
05 /**
06  * Memory of 4K 16-bit registers.
07  * If load is asserted, the value of the register selected by
08  * address is set to in; Otherwise, the value does not change.
09  * The value of the selected register is emitted by out.
10  */
11 CHIP RAM4K {
12   IN in[16], load, address[12];
13   OUT out[16];
14
15   PARTS:
16   DMux8Way(in=load, sel=address[9..11], a=load0, b=load1, c=load2, d=load3, e=load4, f=load5, g=load6,
17   h=load7);
18   RAM512(in=in, load=load0, address=address[0..8], out=ram0);
19   RAM512(in=in, load=load1, address=address[0..8], out=ram1);
20   RAM512(in=in, load=load2, address=address[0..8], out=ram2);
21   RAM512(in=in, load=load3, address=address[0..8], out=ram3);
22   RAM512(in=in, load=load4, address=address[0..8], out=ram4);
23   RAM512(in=in, load=load5, address=address[0..8], out=ram5);
24   RAM512(in=in, load=load6, address=address[0..8], out=ram6);
25   RAM512(in=in, load=load7, address=address[0..8], out=ram7);
26
27   Mux8Way16(a=ram0, b=ram1, c=ram2, d=ram3, e=ram4, f=ram5, g=ram6, h=ram7, sel=address[9..11],
28   out=out);
29 }
```

File: RAM4K(1).hdl | Chip: RAM4K

6. RAM4K — Simulation Screenshot

The screenshot shows the HDL/Chip Simulator interface for the RAM4K chip. The left pane displays the HDL code, and the right pane shows the chip's internal state.

HDL Code (Left Pane):

```
1 .le is part of www.nand2tetris.org
2 : book "The Elements of Computing Systems"
3 in and Schocken, MIT Press.
4 me: projects/3/b/RAM4K.hdl
5
6 of 4K 16-bit registers.
7 ! is asserted, the value of the register selected by
8 : is set to in; Otherwise, the value does not change.
9 ue of the selected register is emitted by out.
10
11 {
12 :16, load, address[12];
13 t[16];
14
15
16 ay(in=load, sel=address[9..11], a=load0, b=load1, c=load2, d=load3, e=load4, f=load5, g=load6, h=load7)
17
18 !(in=in, load=load0, address=address[0..8], out=ram0);
19 !(in=in, load=load1, address=address[0..8], out=ram1);
20 !(in=in, load=load2, address=address[0..8], out=ram2);
21 !(in=in, load=load3, address=address[0..8], out=ram3);
22 !(in=in, load=load4, address=address[0..8], out=ram4);
23 !(in=in, load=load5, address=address[0..8], out=ram5);
24 !(in=in, load=load6, address=address[0..8], out=ram6);
25 !(in=in, load=load7, address=address[0..8], out=ram7);
26
27 y16(a=ram0, b=ram1, c=ram2, d=ram3, e=ram4, f=ram5, g=ram6, h=ram7, sel=address[9..11], out=out);
28
```

Chip RAM4K (Right Pane):

Input pins:

- in: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
- load: 0
- address: 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Output pins:

- out: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Internal pins:

- load0: 0
- load1: 0
- load2: 0
- load3: 0
- load4: 0
- load5: 0
- load6: 0
- load7: 0

RAM Array (Visualization):

RAM512	Addr	dec
0	0	0
1	0	0
2	0	0
3	0	0
4	0	0

Provided screenshot from the HDL/chip simulator.

7. RAM16K.hdl

HDL Source Code

```
01 // This file is part of www.nand2tetris.org
02 // and the book "The Elements of Computing Systems"
03 // by Nisan and Schocken, MIT Press.
04 // File name: projects/3/b/RAM16K.hdl
05 /**
06  * Memory of 16K 16-bit registers.
07  * If load is asserted, the value of the register selected by
08  * address is set to in; Otherwise, the value does not change.
09  * The value of the selected register is emitted by out.
10  */
11 CHIP RAM16K {
12   IN in[16], load, address[14];
13   OUT out[16];
14
15   PARTS:
16   DMux4Way(in=load, sel=address[12..13], a=load0, b=load1, c=load2, d=load3);
17
18   RAM4K(in=in, load=load0, address=address[0..11], out=ram0);
19   RAM4K(in=in, load=load1, address=address[0..11], out=ram1);
20   RAM4K(in=in, load=load2, address=address[0..11], out=ram2);
21   RAM4K(in=in, load=load3, address=address[0..11], out=ram3);
22
23   Mux4Way16(a=ram0, b=ram1, c=ram2, d=ram3, sel=address[12..13], out=out);
24 }
```

File: RAM16K(1).hdl | Chip: RAM16K

7. RAM16K — Simulation Screenshot

The screenshot displays the HDL/Chip Simulator interface for the RAM16K chip. The left pane shows the HDL code, and the right pane shows the chip's internal structure and pin configurations.

HDL Code (Left Pane):

```
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/3/b/RAM16K.hdl
5 /**
6  * Memory of 16K 16-bit registers.
7  * If load is asserted, the value of the register selected by
8  * address is set to in; Otherwise, the value does not change.
9  * The value of the selected register is emitted by out.
10 */
11 CHIP RAM16K {
12     IN in[16], load, address[14];
13     OUT out[16];
14
15     PARTS:
16     DMux4Way(in=load, sel=address[12..13], a=load0, b=load1, c=load2, d=load3);
17
18     RAM4K(in=in, load=load0, address=address[0..11], out=ram0);
19     RAM4K(in=in, load=load1, address=address[0..11], out=ram1);
20     RAM4K(in=in, load=load2, address=address[0..11], out=ram2);
21     RAM4K(in=in, load=load3, address=address[0..11], out=ram3);
22
23     Mux4Way16(a=ram0, b=ram1, c=ram2, d=ram3, sel=address[12..13], out=out);
24 }
```

Chip RAM16K (Right Pane):

Input pins:

- in: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 (dec)
- load: 0
- address: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 (dec)

Output pins:

- out: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 (dec)

Internal pins:

- load0: 0
- load1: 0
- load2: 0
- load3: 0
- ram0: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 (dec)
- ram1: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 (dec)
- ram2: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 (dec)
- ram3: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 (dec)

Visualization:

The visualization section shows three RAM4K blocks, each with 4K addresses (0 to 4) and 16-bit outputs (0 to 15). The outputs are currently all 0.

Provided screenshot from the HDL/chip simulator.

8. PC.hdl

HDL Source Code

```
01 // This file is part of www.nand2tetris.org
02 // and the book "The Elements of Computing Systems"
03 // by Nisan and Schocken, MIT Press.
04 // File name: projects/3/a/PC.hdl
05 /**
06  * A 16-bit counter.
07  * if reset(t): out(t+1) = 0
08  * else if load(t): out(t+1) = in(t)
09  * else if inc(t): out(t+1) = out(t) + 1
10  * else out(t+1) = out(t)
11  */
12 CHIP PC {
13   IN in[16], reset, load, inc;
14   OUT out[16];
15
16   PARTS:
17   Inc16(in=prevOut, out=incOut);
18
19   Mux16(a=prevOut, b=incOut, sel=inc, out=mux1Out);
20   Mux16(a=mux1Out, b=in, sel=load, out=mux2Out);
21   Mux16(a=mux2Out, b=false, sel=reset, out=mux3Out);
22
23   Register(in=mux3Out, load=true, out=prevOut, out=out);
24 }
```

File: PC(1).hdl | Chip: PC

8. PC — Simulation Screenshot

HDL ☐ --LBSt Project 3 PC

```
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/3/a/PC.hdl
5 /**
6  * A 16-bit counter.
7  * if      reset(t): out(t+1) = 0
8  * else if load(t):  out(t+1) = in(t)
9  * else if inc(t):   out(t+1) = out(t) + 1
10 * else              out(t+1) = out(t)
11 */
12 CHIP PC {
13     IN in[16], reset, load, inc;
14     OUT out[16];
15
16     PARTS:
17     Inc16(in=prevOut, out=incOut);
18
19     Mux16(a=prevOut, b=incOut, sel=inc, out=mux1Out);
20     Mux16(a=mux1Out, b=in, sel=load, out=mux2Out);
21     Mux16(a=mux2Out, b=false, sel=reset, out=mux3Out);
22
23     Register(in=mux3Out, load=true, out=prevOut, out=out);
24 }
```

Chip PC Eval Reset Clock: 0

Input pins

in 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

reset 0

load 0

inc 0

Output pins

out 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

Internal pins

prevOut 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

incOut 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 dec

mux1Out 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

mux2Out 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

mux3Out 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

Visualization

Register: 0

Provided screenshot from the HDL/chip simulator.